

The screenshot displays an IDE interface with a project explorer on the left and a code editor on the right.

Project Explorer (Left):

- Project
 - Q_1_demo
 - Q_1_abstract
 - Q_1_cla
 - Q_1_cla.class
 - Q_1_nested.java
 - A
 - Q_1_nested
 - Q_1_objectTypeCasting.java
 - Animal
 - Cat
 - Dog
 - Q_1_objectTypeCasting
 - Q_1_Rectangle
 - Q_1_triangle
 - Q_2_Employee
 - Q_2_Employee
 - Q_2_Full_Time_Employee
 - Q_2_Part_Time_Employee
 - Q_3_IMtech
 - Q_3_IMtech_Complex
 - Q_3_IMtech_Integer
 - Q_3_IMtech_Number**
 - Test
 - Q_1_Abstract_test
 - Q_2_Test
 - Q_3_Test
 - .gitignore
 - E6_25MCCE37.iml
- External Libraries
- Scratches and Consoles

Code Editor (Right):

The editor shows the source code for `Q_3_Imtech_Number.java`. The code defines a package and a public interface with several methods. The interface name `Q_3_Imtech_Number` is highlighted in green.

```
1 package Q_3_IMtech;
2
3 public interface Q_3_Imtech_Number { 11 usages 2 implementations
4     public String toString(); 2 implementations
5     public boolean equals_To(Q_3_Imtech_Number num); no usages 2 implementations
6     public int compare_To(Q_3_Imtech_Number num); no usages 2 implementations
7     public Q_3_Imtech_Number add_To(Q_3_Imtech_Number num); no usages 2 implementations
8     public Q_3_Imtech_Number sub_From(Q_3_Imtech_Number num); no usages 2 implementations
9     public Q_3_Imtech_Number muti_With(Q_3_Imtech_Number num ); no usages 2 implementations
10    public Q_3_Imtech_Number divide_By(Q_3_Imtech_Number num); no usages 2 implementations
11 }
12
```

The image shows a screenshot of an IDE interface. On the left is a 'Project' sidebar with a tree view of a Java project. The project structure includes folders like 'Q_1_demo', 'Q_2_Employee', 'Q_3_IMtech', and 'Test'. The 'Q_3_IMtech' folder is expanded, showing files like 'Q_3_IMtech_Complex', 'Q_3_IMtech_Integer', and 'Q_3_IMtech_Number'. The 'Q_3_IMtech_Integer' file is selected. The main editor area on the right displays the code for 'Q_3_IMtech_Integer.java'. The code is in Java and includes package declarations, class declarations, and method implementations. Line 8, 'this.num = 0;', is highlighted in blue. The bottom status bar shows the file path 'E6_25MCCE37 > src > Q_3_IMtech > Q_3_IMtech_Integer' and the current time '8:21'.

The image shows a screenshot of an IDE interface. On the left is a sidebar with a 'Project' view showing a tree of files and folders. The selected file is 'Q_3_Imtech_Integer.java'. The main editor area displays the code for this file. The code is in Java and includes a class definition, a countDigits method, a checkPrime method, a getDigit method, an is_prime method, and an overridden toString method. The IDE has a top toolbar with various icons for file operations, a bottom status bar showing file encoding and line length, and a taskbar at the very bottom with application icons.

The image shows a screenshot of an IDE with a dark theme. On the left is the Project Explorer showing a project named 'E6_25MCCE37'. It contains several packages: 'Q_1_demo', 'Q_2_Employee', 'Q_3_IMtech', and 'Test'. The 'Q_3_IMtech' package is expanded, showing 'Q_3_IMtech_Complex', 'Q_3_Imtech_Integer' (which is selected), and 'Q_3_Imtech_Number'. The main editor area displays the code for 'Q_3_Imtech_Integer.java'. The code defines a class that implements 'Q_3_Imtech_Number'. It includes an '@Override' method 'equals_To' which compares the 'num' field of the current object with the 'num' field of the passed object. It also includes an 'add_To' method that takes a 'Q_3_Imtech_Number' and returns a new 'Q_3_Imtech_Integer' with the sum of the 'num' fields. The bottom status bar shows the current file path: 'E6_25MCCE37 > src > Q_3_IMtech > Q_3_Imtech_Integer > Q_3_Imtech_Integer'. The system tray at the bottom right shows the time as 12:12 AM on 17/09/26.

The image shows a screenshot of an IDE interface. On the left is a Project Explorer showing a project named 'E6_25MCCE37'. It contains several packages: 'Q_1_demo', 'Q_2_Employee', and 'Q_3_IMtech'. The 'Q_3_IMtech' package is expanded, showing 'Q_3_IMtech_Complex', 'Q_3_Imtech_Integer' (which is selected), and 'Q_3_Imtech_Number'. There is also a 'Test' package with 'Q_1_Abstract_test', 'Q_2_Test', and 'Q_3_Test'. At the bottom of the Project Explorer are '.gitignore' and 'E6_25MCCE37.iml'. The main editor area shows the code for 'Q_3_Imtech_Integer.java'. The code defines a public class that implements 'Q_3_Imtech_Number'. It includes methods 'add_To', 'sub_From', 'muti_With', and 'divide_By'. The 'add_To' method has a warning icon and '20 usages'. The 'muti_With' method has a warning icon and 'no usages'. The 'divide_By' method has a warning icon and 'no usages'. The code is written in Java and uses standard syntax for classes, methods, and variables. The IDE's status bar at the bottom shows the current file path: 'E6_25MCCE37 > src > Q_3_IMtech > Q_3_Imtech_Integer > Q_3_Imtech_Integer'. The system tray at the very bottom shows the time as 8:21, the file encoding as UTF-8, and the number of spaces as 4.

The screenshot displays an IDE interface with a project explorer on the left and a code editor on the right.

Project Explorer (Left):

- Project: E6_25MCCE37
- Q_1_demo
 - Q_1_abstract
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 - Q_1_cla.class
- Q_1_nested.java
 - A
 - Q_1_nested
- Q_1_objectTypeCasting.java
 - Animal
 - Cat
 - Dog
 - Q_1_objectTypeCasting
- Q_1_Rectangle
- Q_1_triangle
- Q_2_Employee
 - Q_2_Employee
 - Q_2_Full_Time_Employee
 - Q_2_Part_Time_Employee
- Q_3_IMtech
 - Q_3_IMtech_Complex** (selected)
 - Q_3_Imtech_Integer
 - Q_3_Imtech_Number
- Test
 - Q_1_Abstract_test
 - Q_2_Test
 - Q_3_Test
- .gitignore
- E6_25MCCE37.iml
- External Libraries
- Scratches and Consoles

Code Editor (Right):

File: Q_3_IMtech_Complex.java

```
1 package Q_3_IMtech;
2
3 public class Q_3_IMtech_Complex implements Q_3_Imtech_Number { 22 usages
4
5     int real; 24 usages
6     int imag; 25 usages
7
8     public Q_3_IMtech_Complex(int real, int imag) { 6 usages
9         this.real = real;
10        this.imag = imag;
11    }
12
13    public double getMag() { 4 usages
14        return Math.sqrt(real * real + imag * imag);
15    }
16
17    public int getReal() { no usages
18        return real;
19    }
20
21    public int getImag() { no usages
22        return imag;
23    }
24
25    public String toString() {
26
27        if (imag >= 0) {
28            return real + " + " + imag + "i";
29        }
30        else {
31            return real + " - " + (-imag) + "i";
32        }
33    }
```

Bottom Bar:

E6_25MCCE37 > src > Q_3_IMtech > Q_3_IMtech_Complex > getReal

19:6 LF UTF-8 4 spaces

12:13 AM 17/09/26

The image shows a screenshot of an IDE, likely IntelliJ IDEA, with a dark theme. The top bar displays the project name 'E6_25MCCE37' and 'Version Control'. The left sidebar shows the 'Project' view with a tree structure of files and folders. The main editor area shows the code for 'Q_3_IMtech_Complex.java', which implements the 'Q_3_IMtech_Number' interface. The code includes methods for getting real and imaginary parts, converting to a string, and comparing with another complex number. The bottom status bar shows the current file path: 'E6_25MCCE37 > src > Q_3_IMtech > Q_3_IMtech_Complex > getReal'.

```
Employee.java  Q_2_Test.java  Q_3_Imtech_Number.java  Q_3_Imtech_Integer.java  Q_3_IMtech_Complex.java x  v  :  🔔
3      public class Q_3_IMtech_Complex implements Q_3_Imtech_Number { 22 usages  ⚠️ 3  ✅ 1  ^  v
41
42  📌  public int compare_To(Q_3_Imtech_Number n) { no usages
43
44      Q_3_IMtech_Complex other = (Q_3_IMtech_Complex) n;
45
46      if (getMag() < other.getMag()) {
47          return -1;
48      }
49      else if (getMag() == other.getMag()) {
50          return 0;
51      }
52      else {
53          return 1;
54      }
55  }
56
57  📌  public Q_3_Imtech_Number add_To(Q_3_Imtech_Number n) { no usages
58
59      if (n instanceof Q_3_IMtech_Complex) {
60
61          Q_3_IMtech_Complex other = (Q_3_IMtech_Complex) n;
62
63          return new Q_3_IMtech_Complex(
64              real: real + other.real,
65              imag: imag + other.imag
66          );
67      }
68
69      else if (n instanceof Q_3_Imtech_Integer) {
70
71          Q_3_Imtech_Integer other = (Q_3_Imtech_Integer) n;
72
73          return new Q_3_IMtech_Complex(
74              real: real + other.getReal(),
75              imag: imag + other.getImag()
76          );
77      }
78  }
```


The image shows a screenshot of an IDE, likely IntelliJ IDEA, with a dark theme. The top bar displays the project name 'E2 E6_25MCCE37' and 'Version Control'. The left sidebar shows a 'Project' view with a tree structure. The tree is expanded to show the 'Q_3_IMtech' package, which contains 'Q_3_IMtech_Complex', 'Q_3_IMtech_Integer', and 'Q_3_IMtech_Number'. The 'Q_3_IMtech_Complex' class is selected. The main editor area shows the code for 'Q_3_IMtech_Complex.java'. The code defines a class that implements 'Q_3_IMtech_Number' and contains three methods: 'add_To', 'sub_From', and 'muti_With'. The 'add_To' method is currently selected and highlighted. The code is as follows:

```
3 public class Q_3_IMtech_Complex implements Q_3_IMtech_Number { 22 usages
57     public Q_3_IMtech_Number add_To(Q_3_IMtech_Number n) { no usages
69         else if (n instanceof Q_3_IMtech_Integer) {
70
71             Q_3_IMtech_Integer other = (Q_3_IMtech_Integer) n;
72
73             return new Q_3_IMtech_Complex(
74                 real: real + other.num,
75                 imag
76             );
77         }
78
79         return null;
80     }
81
82     public Q_3_IMtech_Number sub_From(Q_3_IMtech_Number n) { no usages
83
84         Q_3_IMtech_Complex other = (Q_3_IMtech_Complex) n;
85
86         return new Q_3_IMtech_Complex(
87             real: real - other.real,
88             imag: imag - other.imag
89         );
90     }
91
92     public Q_3_IMtech_Number muti_With(Q_3_IMtech_Number n) { no usages
93
94         Q_3_IMtech_Complex other = (Q_3_IMtech_Complex) n;
95
96         int r = real * other.real - imag * other.imag;
97
98         int i = real * other.imag + imag * other.real;
99
100         return new Q_3_IMtech_Complex(r, i);
```

 The bottom status bar shows the file path 'E6_25MCCE37 > src > Q_3_IMtech > Q_3_IMtech_Complex > getReal' and the current line '19:6 LF UTF-8 4 spaces'.

The image shows a screenshot of an IDE interface. On the left is a 'Project' sidebar with a tree view of a Java project. The project structure includes folders like 'Q_1_demo', 'Q_2_Employee', and 'Q_3_IMtech'. The 'Q_3_IMtech' folder is expanded, showing files 'Q_3_IMtech_Complex', 'Q_3_Imtech_Integer', and 'Q_3_Imtech_Number'. The 'Q_3_IMtech_Complex' file is selected. The main editor area on the right displays the code for 'Q_3_IMtech_Complex.java'. The code defines a class that implements 'Q_3_Imtech_Number' and includes methods for multiplication ('muti_With') and division ('divide_By'). The code is written in Java and uses color-coded syntax. The bottom status bar shows the current file path: 'E6_25MCCE37 > src > Q_3_IMtech > Q_3_IMtech_Complex > getReal'. The system tray at the very bottom shows the time as 12:13 AM on 17/09/26.

The image shows a screenshot of an IDE, likely IntelliJ IDEA, with a dark theme. The top bar displays the project name 'E2 E6_25MCCE37' and 'Version Control'. The left sidebar shows the 'Project' view with a tree structure of files and folders. The main editor area shows the code for 'Q_3_Test.java'. The code is a Java program that tests various methods of the 'Q_3_IMtech' package, including 'Integer', 'Complex', and 'Number'. The code is color-coded, and line numbers are visible on the left of the editor. The bottom status bar shows the current file path 'E6_25MCCE37 > src > Test > Q_3_Test > main' and the time '19:64'.


```

/usr/lib/jvm/default/bin/java -javaagent:/usr/share/idea/lib/idea_rt.jar=37263 -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding=UTF-8 -classpath /home/va
INTEGER
Number 1 = 25
Number 2 = 5
Digit at position 0 = 5
Digit at position 1 = 2
Is 25 prime? = false
Equal = false
Compare = 1
Addition = 7
Subtraction = 7
Multiplication = 10
Division = 0

Complex 1 = 3 + 4i
Complex 2 = 1 + 2i
Real part = 3
Imaginary part = 4
Magnitude = 5.0
Equal = false
Compare = 1
Addition = 4 + 6i
Subtraction = 2 + 2i
Multiplication = -5 + 10i
Division = 2 + 0i
MIXED ADDITION
Integer + Complex = 5 + 4i
Complex + Integer = 5 + 4i

Process finished with exit code 0

```